

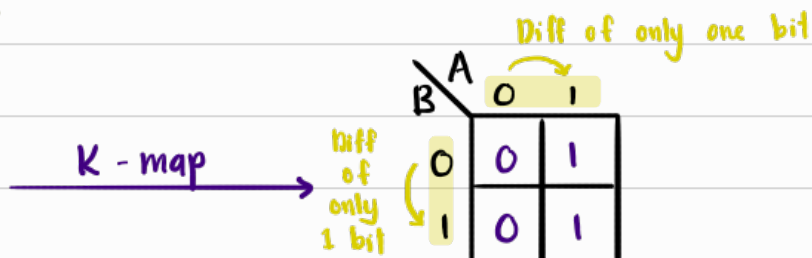
Karnaugh Maps [K map]

~ Special form of a truth table which enables easier pattern recognition

let's say we have these truth tables

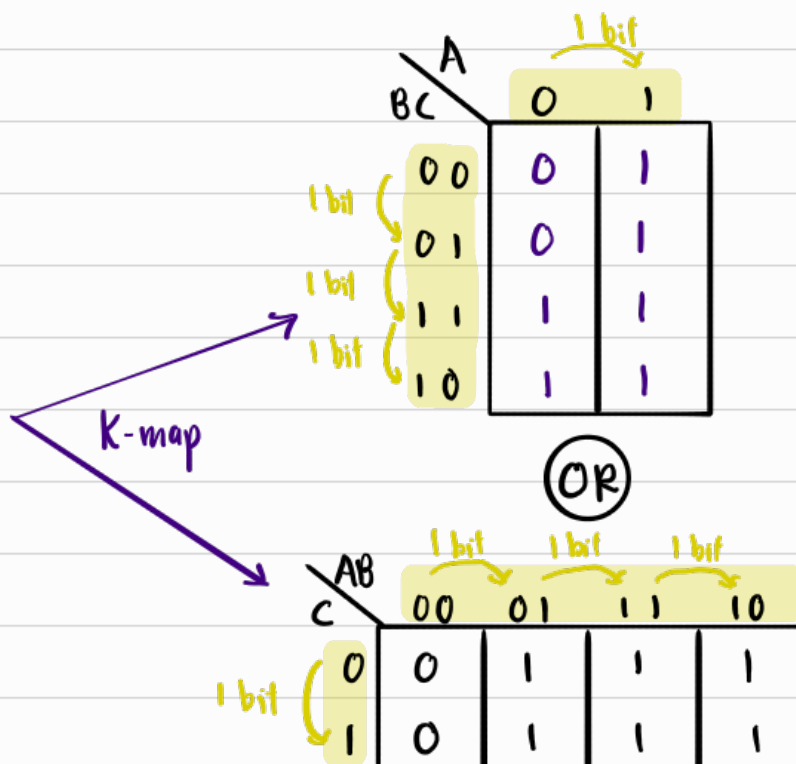
2 variables

A	B	F
0	0	0
0	1	0
1	0	1
1	1	1



3 variables

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1



Deriving Expressions from a Karnaugh Map [Sum of product]

① Find the largest group of 1 [No 0]

↳ Grouping rules ...

1) horizontal or vertical [Squares or Rectangles]

2) Contain 2^n # of 1s (1, 2, 4, 8 ...)

3) As large as possible

4) Can overlap

5) Can wrap around a table

6) All 1s must be in a grp

② Compare & observe dependent & independent variables

↳ 1 group, 1 result(s) [Result can be none]

↳ Expression, $X = (\text{result})_1 + (\text{result})_2 + \dots$

• However, if within 1 group have > 1 dependent variables

↳ Expression $X = AB\dots$ [Example 6, Example 8]

Shortcut for step 1 [Finding dependent & independent variables/input]

Within a group ...

Since non-common variable where the output is always 1, guarantees the output is independent of that variable

① Ignore non-common variable

② Is it "not"?

↳ If variable is 1, its A

↳ If variable is 0, its $\bar{A}$

Take note:

This method of grouping 1s is called **sum of products**, becz the final expression is gotten from summing up the products of each respective groups of 1.

An alternative method is called **product of sums**. Final expression is gotten from multiplying up all the sum of each respective grp of 0.

2 Variables

Example 1

When $A = 1$, $Out = 1$

A \ B	0	1
0	0	1
1	0	1

← When $B = 0$, $Out = 1$

← When $B = 1$, $Out = 1$

} This means whatever B is, $out = 1$
 $\therefore$ Output is independent of B

$\therefore$ Expression is $X = A$

Example 2

		A	
		0	1
B	0	0	1
	1	1	1

$A = 1, Out = 1$
 $B = 1, out = 1$

$\leftarrow B = 0, Out = 1$
 $\leftarrow B = 1, Out = 1$

$\uparrow \quad \uparrow$
 $A = 0, A = 1,$
 $Out = 1 \quad Out = 1$

Output independent of A

Output independent of B

$\therefore$ Expression is $X = A + B$

Example 3

		$\bar{A}$	
		0	1
B	0	1	1
	1	1	0

$A = 0, Out = 1$
 $B = 0, out = 1$

$\leftarrow B = 0, Out = 1$
 $\leftarrow B = 1, Out = 1$

$\uparrow \quad \uparrow$
 $A = 0, A = 1,$
 $Out = 1 \quad Out = 1$

Output independent of A

Output independent of B

$\therefore$ Expression is $X = \bar{A} + \bar{B}$

3 Variables

Example 4

Truth table for Example 4:

BC		A	
		0	1
0	0	0	1
0	1	0	1
1	1	1	1
1	0	1	1

Annotations:

- $A = 1, Out = 1$ (purple box)
- $B = 1, Out = 1$ (green box)
- $C = 1, Out = 1$ (green box)
- $C = 0, Out = 1$ (green box)
- $A = 0 \text{ or } 1, Out = 1$ (green box)
- Output independent of B (yellow text)
- Output independent of C (yellow text)
- Output independent of A (yellow text)

$\therefore X = A + B$

Example 5

Truth table for Example 5:

C	AB			
	00	01	11	10
0	1	1	1	1
1	0	0	1	1

Annotations:

- $\bar{C} = 1, Out = 1$ (purple box)
- $A = 0 \text{ or } 1, Out = 1$ (purple box)
- $B = 0 \text{ or } 1, Out = 1$ (purple box)
- $C = 0 \text{ or } 1, Out = 1$ (green box)
- $A = 1, Out = 1$ (green box)
- $B = 0 \text{ or } 1, Out = 1$ (green box)
- Output independent of A (yellow text)
- Output independent of B (yellow text)
- Output independent of C (yellow text)

$\therefore X = A + \bar{C}$

Example 6

Truth table for Example 6:

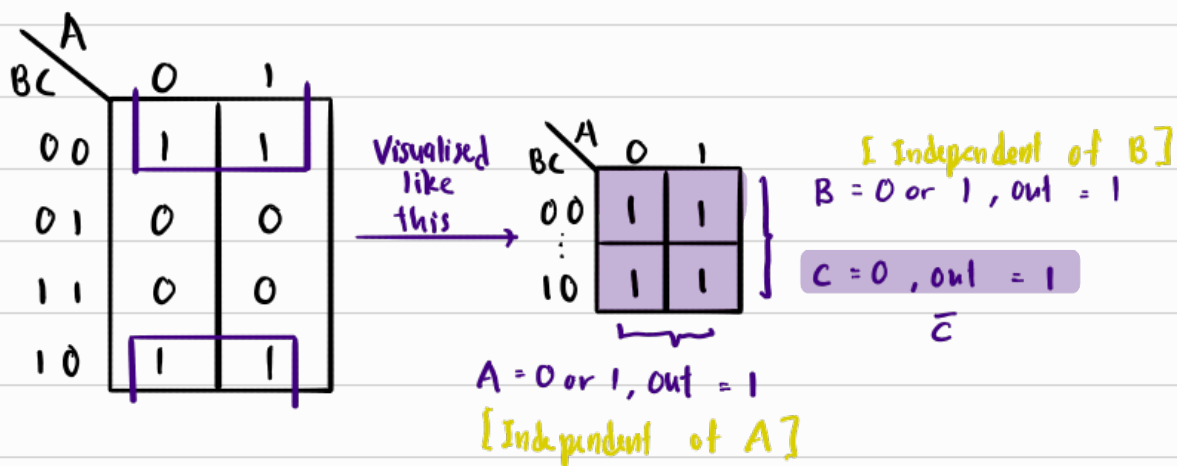
C	AB			
	00	01	11	10
0	0	0	1	0
1	0	0	1	0

Annotations:

- $A = 1, Out = 1$ (purple box)
- $B = 1, Out = 1$ (purple box)
- $C = 0 \text{ or } 1, Out = 1$ (purple box)
- Output independent of C (yellow text)

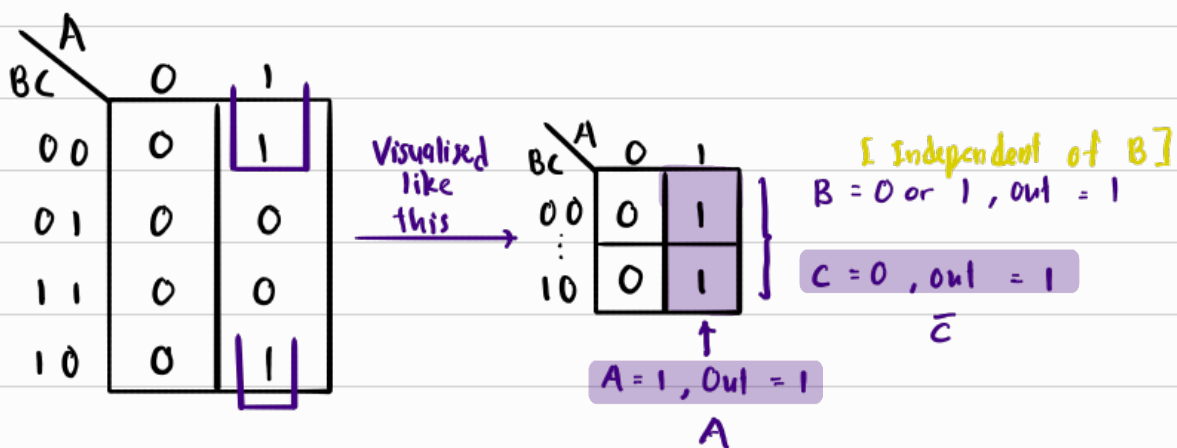
$\therefore X = AB$

Example 7



$$\therefore X = \bar{C}$$

Example 8



$$\therefore X = A\bar{C}$$

Karnaugh Map Terminology

Term	Definition	Example (3 variables: A, B, C)	In K-map
Minterm	A product (AND) term where all variables appear once (complemented or not). Represents a single 1-cell in the map.	$A'B'C \rightarrow$ true only when A=0, B=0, C=1	One cell marked 1
Maxterm	A sum (OR) term where all variables appear once (complemented or not). Represents a single 0-cell in the map.	$(A + B + C') \rightarrow$ false only when A=0, B=0, C=1	One cell marked 0
Implicant	A product term that covers one or more minterms. In the map, any group of 1s forming a rectangle of size 2^n .	Grouping $AB'C$ and $AB'C' \rightarrow$ implicant = AB'	A block of 1s
Prime Implicant	An implicant that cannot be combined into a larger group. (The largest possible block of 1s.)	A 4-cell group $\rightarrow A$	Biggest rectangle of 1s
Essential Prime Implicant	A prime implicant that covers at least one unique minterm not covered by any other. Must be included in final simplification.	If only one block covers $A'B'C$, that block is essential	Highlighted must-have block
Redundant Implicant	An implicant that doesn't add coverage (all its 1s already covered by others). Can be omitted.	A 2-cell group fully inside a 4-cell group	Extra, unnecessary grouping
Don't-care (X)	Input combos that are irrelevant or impossible. Can be treated as 0 or 1 to aid simplification.	$XYZ=111$ never happens $\rightarrow$ mark cell as X	Flexible cell

Canonical Form

K-maps above 5 variables are extremely time consuming to do.

another method is to convert truth tables back into a boolean expression and use theorems to simplify it. This is called the Canonical Form.

For illustration purpose, let's take the truth table of example 2

A	B	F
0	0	0
0	1	1
1	0	1
1	1	1

There are 2 ways to represent this table

- ① List out all the combination that gives true
- ② List out all the combination that gives false

SOP

Method ① [Sum of Products / Sum of Minterms]

$$\begin{array}{lcl} F = 1 \text{ when } A = 0, B = 1 & \longrightarrow & A'B \\ & & A = 1, B = 0 \longrightarrow AB' \\ & & A = 1, B = 1 \longrightarrow AB \end{array} \left. \vphantom{\begin{array}{l} F = 1 \text{ when } A = 0, B = 1 \\ A = 1, B = 0 \\ A = 1, B = 1 \end{array}} \right\} \begin{array}{l} \text{Minterms} \\ \text{Product of variables} \\ \text{that gives true [1]} \end{array}$$

$$\begin{aligned} \therefore X &= A'B + AB' + AB \quad \longleftarrow \text{This expression is true and only true} \\ &= A'B + A \quad \text{when any of the minterms is true, that is ...} \\ &= (A' + A) \cdot (A + B) \quad A = 0, B = 1 \text{ OR } A = 1, B = 0 \text{ OR } A = 1, B = 1 \\ X &= A + B \quad \left[\text{This reads logically as, } X \text{ is true when } AB \text{ or } AB' \text{ or } AB \right. \\ &\quad \left. \text{is true} \right] \end{aligned}$$

POS

Method ② [Products of Sum / Product of Maxterms]

$$F = 0 \text{ when } A = 0, B = 0 \longrightarrow A + B \left. \vphantom{F = 0 \text{ when } A = 0, B = 0} \right\} \begin{array}{l} \text{Maxterm} \\ \text{Sum of variable} \\ \text{that give false [0]} \end{array}$$

$$\begin{aligned} \therefore X &= (A + B) \quad \longleftarrow \text{This expression is false and only false} \\ &\quad \text{when any of the maxterms are false, that is ..} \\ &\quad A = 0, B = 0 \\ &\quad \left[\text{This reads logically as, ONLY if NONE of the maxterms} \right. \\ &\quad \left. \text{are false, } X \text{ will be true} \right] \end{aligned}$$

Alternate Canonical Form [Decimal]

	Inputs			Output	Min term	Max term
	p	q	r	f		
000 → 0	0	0	0	0	$m_0 = \bar{p}\bar{q}\bar{r}$	$M_0 = p+q+r$
001 → 1	0	0	1	0	$m_1 = \bar{p}\bar{q}r$	$M_1 = p+q+\bar{r}$
010 → 2	0	1	0	0	$m_2 = \bar{p}q\bar{r}$	$M_2 = p+\bar{q}+r$
011 → 3	0	1	1	1	$m_3 = \bar{p}qr$	$M_3 = p+\bar{q}+\bar{r}$
100 → 4	1	0	0	0	$m_4 = p\bar{q}\bar{r}$	$M_4 = \bar{p}+q+r$
101 → 5	1	0	1	1	$m_5 = p\bar{q}r$	$M_5 = \bar{p}+q+\bar{r}$
110 → 6	1	1	0	1	$m_6 = pq\bar{r}$	$M_6 = \bar{p}+\bar{q}+r$
111 → 7	1	1	1	1	$m_7 = pqr$	$M_7 = \bar{p}+\bar{q}+\bar{r}$

SOP : $X = \sum \overset{\text{min term}}{m}(3, 5, 6, 7)$

POS : $X = \prod \underset{\text{max term}}{M}(0, 1, 2, 4)$

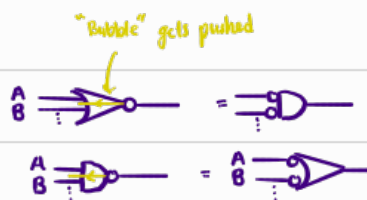
Bubble Pushing

A shortcut method to form equivalent logic circuits based on De Morgan theorem.

De Morgan Theorem

$$\overline{A + B + \dots} = \bar{A} \cdot \bar{B} \cdot \dots$$

$$\overline{A \cdot B \cdot \dots} = \bar{A} + \bar{B} + \dots$$



∴ Pushing 1 bubble from each input to the output gives 1 bubble at the output, but you have to flip the gate [AND ↔ OR] and vice versa

Another important to note is

the involution theorem, $\overline{\overline{X}} = X$

↳ Double bubble = No bubble



∴ This means we can add a double bubble [Cancelling Bubble pair] whenever we want without changing the expression

Example

Convert the following to a and gate



① Since we need at least 1 bubble on each input to push through,

we add a double bubble at B



This technique is really important since NAND gates are universal gates. This method allows us to convert any gate into NAND gate

Circuit diagram generation [Bubble Pushing]

Example 1

Given this truth table ,
implement it using only NAND gates

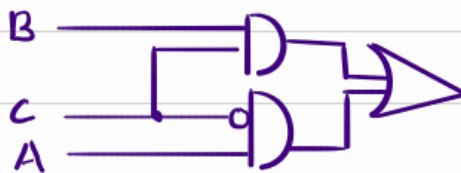
S	A	B	Out
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

① Find the Boolean expression using K-map -

AB \ C	0	1
00	0	0
01	0	1
11	1	1
10	1	0

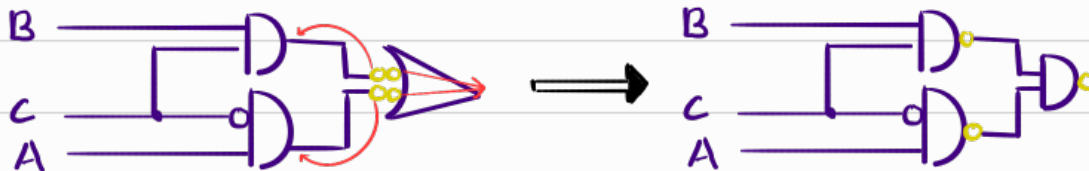
$$\therefore AC' + BC$$

② Draw the circuit according to the Boolean Expression

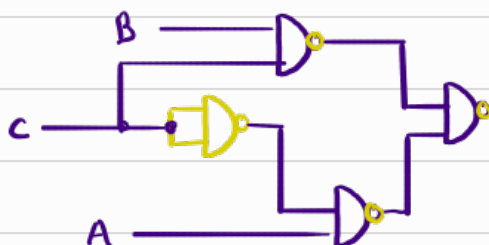


③ Convert all gates to NAND gates starting from output side.

1) Adding 2 double bubbles at OR gate to push



④ Extra bubbles can be converted into a "NAND NOT gate" [X NOT]



Example 2 [Hard]

Given this Half adder truth table, implement it only using NAND gates

A	B	S	Carry'
0	0	0	1
0	1	1	1
1	0	1	1
1	1	0	0

① Find the Boolean expression using K-map -

S

	A	0	1
B	0	0	1
	1	1	0

$$\therefore \text{sop} : A'B + AB'$$

$$\text{POS} : (A+B)(A'+B')$$

$$= (A+B)(\overline{AB})$$

$$= A \cdot \overline{AB} + B \cdot \overline{AB}$$

we choose this
beuz it has $\overline{AB}$
just like Carry'

Carry'

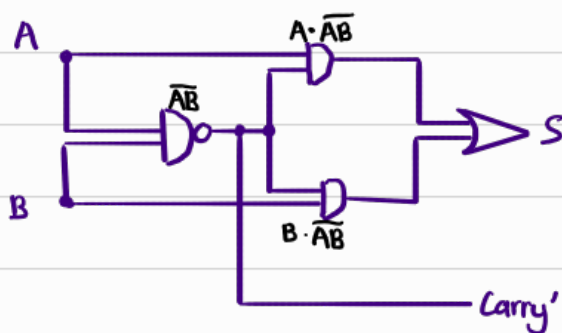
	A	0	1
B	1	1	1
	0	1	0

$$\therefore A' + B' = \overline{AB}$$

Try to change to
NAND to do less
bubble pushing

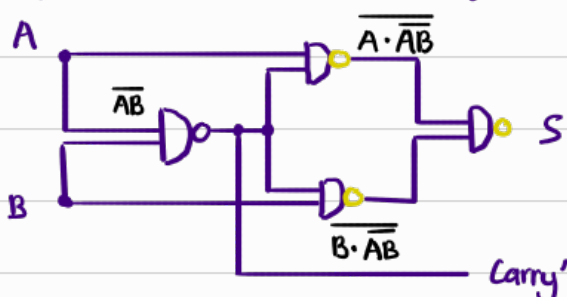
$$\overline{A \cdot \overline{AB} + B \cdot \overline{AB}} = \overline{A \cdot \overline{AB}} \cdot \overline{B \cdot \overline{AB}}$$

② Draw the circuit according to the Boolean Expression



③ Convert all gates to NAND gates starting from output side.

1) Adding 2 double bubble at OR gate



Alternate Method [Brute Force]

$$\begin{aligned} \textcircled{1} \quad S &= A \cdot \overline{AB} + B \cdot \overline{AB} \\ &= \overline{\overline{A \cdot \overline{AB}} + \overline{B \cdot \overline{AB}}} \\ &= \overline{A \cdot \overline{AB}} \cdot \overline{B \cdot \overline{AB}} \quad \# \end{aligned}$$

$$\text{Carry}' = \overline{AB}$$

Egn with no "~+", only "~."

So we can easily draw the diagram

